# Power Series Solutions to Nonlinear Ordinary Differential Equations
# MATH 689
# Homework 8

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April 15, 2026

## 1. Eulerization of the meniscuss at a flat wall in an infinite liquid pool [1], described by the ODE

$$\frac{dh}{dx} = -2\left[(h^2-2)^{-2} - \frac{1}{4}\right]^{1/2}, \quad h(x=0) = \sqrt{2(1-\sin\theta)}, \quad x \in [0, \infty) \tag{1}$$

where $h(x)$ is the meniscus height at a distance $x$ away from the wall. We took $\theta = \pi/4$ and also will do that here.

**(a)** The initial plot from extending the domain to $x \in [0, 5]$ yields

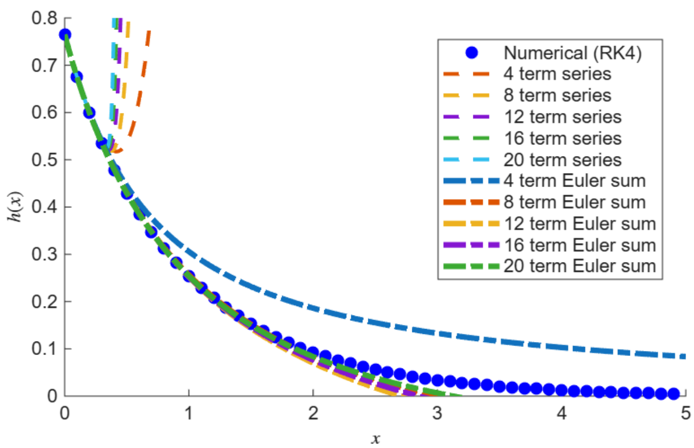



Figure 1: Initial Extension

**(b)** From Dominant Balance, the asymptotic behavior about $x \to \infty$ is

$$h \sim Ce^{-x} + \frac{3}{16}C^3e^{-3x} + \frac{25}{256}C^5e^{-5x} + \cdots, \quad x \to \infty$$

which follows a pattern like

$$h = \exp(-x) \sum_{0 \le n \le \infty} b_n(\exp(-2x))^n \tag{2}$$

If we define a dependent variable transform a

$$h(x) = \exp(-x)H(g(x)) \tag{3}$$

and an independent variable transform

$$g(x) = \exp(-2x), \tag{4}$$

we can write (2) as a standard power series in g:

$$H = \sum_{0 \le n \le \infty} b_n g^n \tag{5}$$

Our goal is to find the $b_n$ coefficients above by transforming (1) into an ODE written in terms of $H$ and $g$ and then find the power series solution (5). We seek a recursion for the $b_n$ coefficients. Let $b_0 = C \approx 0.68333$ for $\theta = \pi/4$.

**(c)**. We substitute (3) into (1) to show (6.2), with the first derive and initial condition

$$\frac{dh}{dx} = -e^{-x}H + e^{-x}H', \qquad H(x=0) = \sum_{0\le n\le\infty} b_n = h(x=0)e^0 = \sqrt{2-\sqrt{2}} \tag{6.1}$$

Then direction substitution yields the form

$$e^{-x}\frac{dH}{dx} - e^{-x}H = -2\left[(e^{-2x}H^2-2)^{-2} - \frac{1}{4}\right]^{1/2} \tag{6.2}$$

**(d)** Moreover, we can write the chain rule as

$$\frac{dH}{dx} = \frac{dH}{dg}\frac{dg}{dx} = \frac{dH}{dg}\left(-2e^{-2x}\right) = \frac{dH}{dg}\left(-2g\right) \tag{7}$$

**(e)** It follows we write (6.2) using (7) and (4)

$$-2\sqrt{g}(g)\frac{dH}{dg} - \sqrt{g}H = -2\left[(gH^2-2)^{-2} - \frac{1}{4}\right]^{1/2}, \quad H(g=0) = b_0 + \sum_{1\le n\le\infty} b_n(0)^n = C \tag{8}$$

**(f)** Equation (8) can be simplified

$$\left(2g\frac{dH}{dg} + H\right)(2-gH^2) = H\sqrt{4-gH^2}, \quad H(g=0) = C,$$

with a power series solution given by (5) with coefficients as

$$\begin{aligned} b_{n>0} &= \frac{1}{4n}\sum_{0\le j\le n-1} b_j\left[(1+2j)B_{n-j-1} + D_{n-j}\right], \quad b_0 = C, \\ D_{n>0} &= \frac{1}{4n}\sum_{1\le j\le n}\left(n - \frac{3}{2}j\right)B_{j-1}D_{n-j}, \quad D_0 = 2, \\ B_n &= \sum_{0\le j\le n} b_j b_{n-j} \end{aligned} \tag{9}$$

Adding the indirect re-summation given by (2) and (9) yields

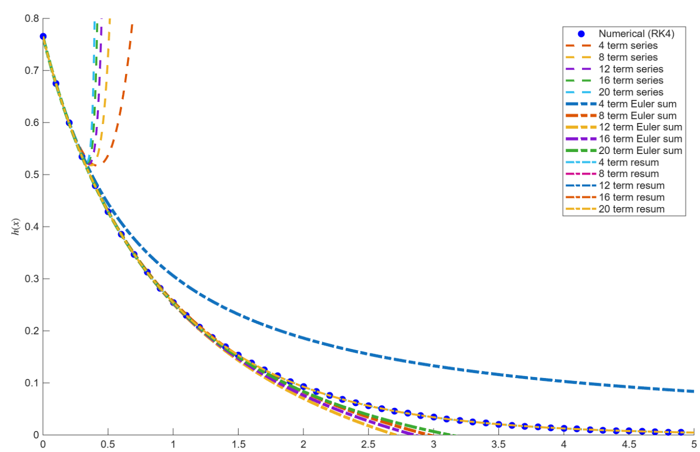



Figure 2: Resummation via the gauge $g = e^{-2x}$

If we let $N = 4$ and plot the resummation via the gauge $g$ we can notice that at the window scale for $x \in [0, 5]$, the solution is accurate to the naked eye ONLY.

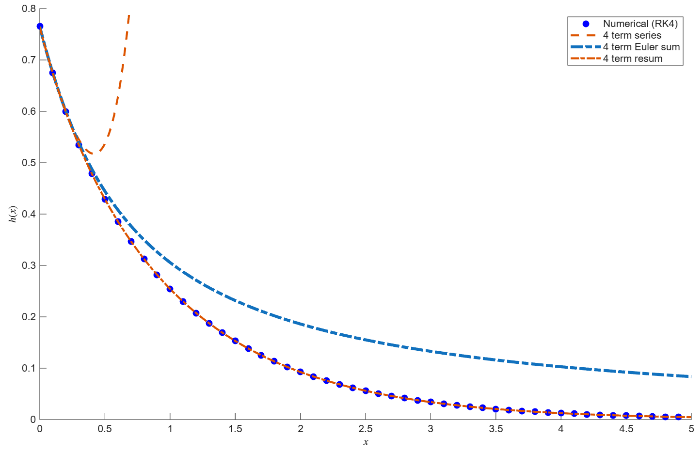



Figure 3: Resummation for $N = 4$

## 2. Independent Variable Transform

Given the ODE

$$\frac{df}{dx} + f = \frac{1}{1+x}, \quad f(x=0) = 1$$

Then we follow with the dependent variable transform $f = F(g(x))$ and independent variable transform $g(x) = x/(1+x)$ so that the above will be written as $F$ and $g$. We can easily show this with first solving for the inverse function "$x(g)$"

$$g + gx(g) = x(g) \Rightarrow x(g) = \frac{g}{1-g}, \quad \frac{dg}{dx} = \frac{(1+x)-x}{(1+x)^2} = \frac{1}{(1+x)^2}$$
$$\frac{df}{dx} = \frac{dF}{dg}\frac{dg}{dx} = \frac{dF}{dg}\frac{1}{(1+x)^2}$$

Then we perform the inverse dependent variable substitution as $F(g) = f(x(g)) \to f(x) = F(g(x))$ using the pieces derived above

$$\frac{dF}{dg}\frac{dg}{dx} + F = \frac{1}{1+x} \Rightarrow \frac{dF}{dg}\frac{1}{(1+x)^2} + F = \frac{1}{1+x}$$
$$\frac{dF}{dg}\frac{1}{(1+(g/(1-g))^2} + F = \frac{1}{1+(g/(1-g))} \Rightarrow \frac{dF}{dg}(1-g)^2 + F = 1-g$$

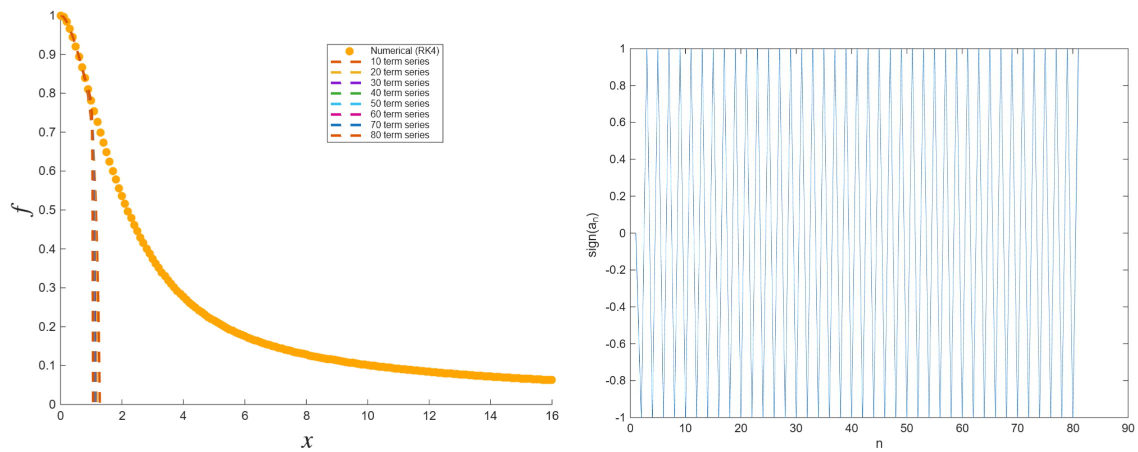
Dividing by $(1-g)^2$ across yields

$$\frac{dF}{dg} + \frac{F}{(1-g)^2} = \frac{1}{1-g}, \quad F(g=0) = 1 \tag{10}$$

The initial conditions (10) arises from the fact that $g = 0 = \frac{x}{1+x} \Rightarrow x = 0$, then note the substitution $f(x=0) = F(g(0) = 0) = 1$.

## 3. MATH 689 Problem

**(a)** We can see that the series coefficients alternate from `sign(a)` and we plot sign($a_n$) for visualization




(a) Initial Plot of Domain Extension with $N = 80$ (b) sign $(a_n)$

Figure 4: Motivation for Eulerization

This motivates the series resummation through Eulerization.
**(b)** The color order index replacement is shown in the plot in (c).
**(c)** The first implmentation of the Eulerization can be observed with worse performance as $N$ increases.

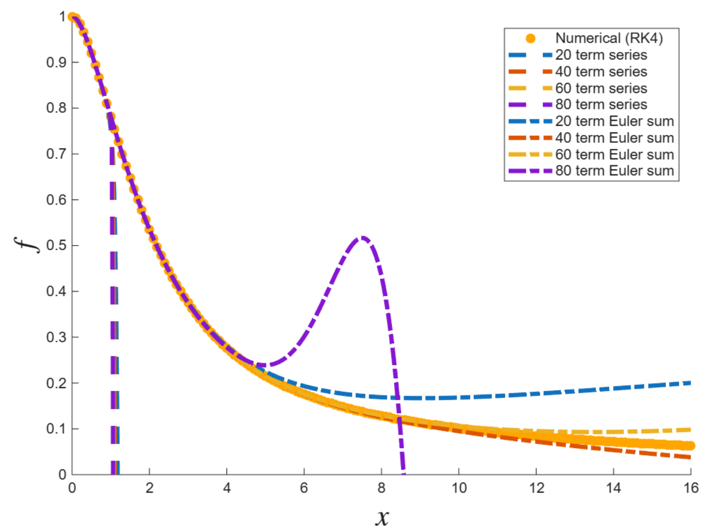



Figure 5: Eulerization

**(d)** We can perform Eulerization by finding the power series solution to the transformed ODE in Problem (2) to overcome the round-off error.

Suppose there exist a power series solution via Eulerization for (10) as

$$F = \sum_{0 \le n \le \infty} A_n g^n, \quad f = \sum_{0 \le n \le \infty} A_n \left( \frac{x}{1+x} \right)^n \tag{A.1}$$

From substitution of (A.1) into (10)

$$\sum_{0 \le n \le \infty} (n+1) A_{n+1} g^n + \left( \sum_{0 \le n \le \infty} A_n g^n \right) \left( \sum_{0 \le n \le \infty} (n+1) g^n \right) = \sum_{0 \le n \le \infty} g^n \tag{A.2}$$

With some algebraic arrangement and Cauchy product yields

$$
\begin{aligned}
\sum_{0\le n\le\infty}(n+1)A_{n+1}g^n+\sum_{0\le n\le\infty}\left(\sum_{0\le j\le n}A_j(n-j+1)\right)g^n &= \sum_{0\le n\le\infty}g^n \\
\sum_{0\le n\le\infty}\left[(n+1)A_{n+1}+\sum_{0\le j\le n}A_j(n-j+1)\right]g^n &= \sum_{0\le n\le\infty}g^n
\end{aligned}
\tag{A.3}
$$

The series recurrence is then given by

$$
A_{n+1}=\frac{1-\left(\sum_{0\le j\le n}A_j(n-j+1)\right)}{n+1},\quad A_0=F(g=0)=1,\quad A_1=\frac{1-A_0}{1}=0 \tag{A.4}
$$

**(e)** We plot the power series solution to (A.4)

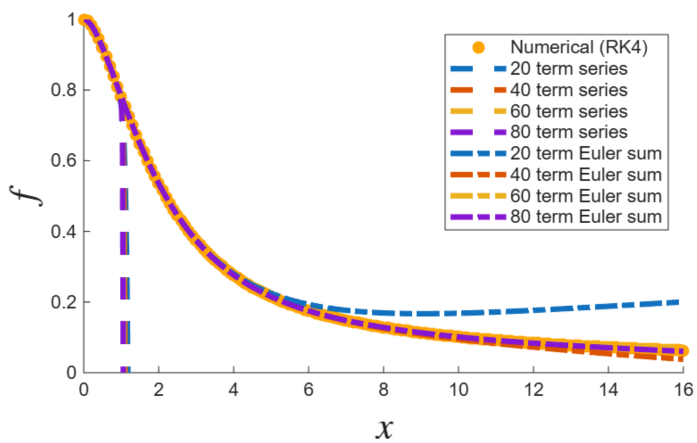



Figure 6: Plot of the Power Series